MULTIPLEX DEVELOPMENT TOOLS

Catalog

Advanced Vehicle Technology
Electrical/Electronic Systems Engineering
Multiplex Standards & Applications

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1.0 General Products

Several products are available that allow a user to design, test, and simulate vehicle multiplexed networks. These tools are presented below.

1.1 Multiplex Reference DataBase (MRDB)

The Multiplex Reference DataBase (MRDB) is used to store information regarding network data, diagnostic data, and vehicle-network data. More precisely, the MRDB contains several relational databases that are used to store information about nodes, functionally addressed messages, physically addressed messages, PIDs, commands, response codes, format codes, DTCs, and vehicle configurations. The MRDB is used in conjunction with several tools that monitor, communicate with, and diagnose networks and modules. This tool is normally provided as part of the Multiplex Development System.

In addition to the MRDB, a primitive-level library of routines is provided that allows users to access the database. The primitive-level library consists of DOS and Windows® compatible "C" routines that allow users to design custom applications which interface to the MRDB.

Product Contents:

- MRDB Files (Borland® Paradox® Files)
- MRDB Primitive-Level Library (Borland® & Microsoft® compatible)
- User Documentation
- *Part of the Multiplex Development System*

Requirements:

- 5 MB Hard Disk Space

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS.

Unit Cost: Free

Notes: The MRDB is distributed with the Vehicle Configuration Editor (VCE).

1.2 Vehicle Configuration Editor (VCE)

The Vehicle Configuration Editor (VCE) is a Microsoft® Windows® based tool that allows a user to define a vehicle network architecture. Both, SCP and ACP are supported. Specifically, a user defines the modules contained within a network and the message traffic between those modules. The VCE can then be used to generate reports or to interface to various network simulators and/or network generation tools. This tool is normally provided as part of the Multiplex Development System.

Enhanced functionality includes the ability to define vehicle diagnostics such as the Diagnostic Trouble Codes (DTCs), Commands, and Parameter Identifiers (PIDs) supported by each module of a vehicle. In addition, the VCE can be used to generate input into the SCP and ACP network simulators and the SCP and ACP Message Verification Test Tools.

Product Contents:
- VCE Software
- User Documentation
- Multiplex Reference DataBase (MRDB)
- *Part of the Multiplex Development System*

Requirements:
- Multiplex Reference DataBase (MRDB)
- Microsoft® Windows® 3.x
- 7 MB Hard Disk Space

Available Training:
Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS.

Unit Cost: Free

2.0 SCP Specific Products

Several products are available that apply specifically to SCP networks. These tools allow a user to develop, test, diagnose, and simulate modules on an SCP network. This section details the following tools:

- SCP Network Analyzer Tool (NAT)
- Dual SCP Peripheral Adaptor (Dual SPA / DSPA)
- Network Simulator
- Data Logger
- ECU Test Tool
- Message Verification Test Tool

These tools and their capabilities are presented below in more detail.

2.1 SCP Network Analyzer Tool (NAT)

The Network Analysis Tool (NAT) is a development tool intended for use by design engineers of Ford SCP modules. The NAT has the capability to act as an SCP message generator, message acknowledger, and network monitor. It is especially suited to the early phases of hardware and software development where it is important to have a stable and repeatable source of messages directed to the module under development. In addition, the monitor function allows a quick and concise display of messages appearing on the network. The NAT features a non-volatile memory for storage of set-up parameters and commonly used messages.

A standard "dumb" terminal or PC equipped with terminal emulation software is used as the user interface, at least for set-up purposes, and for network monitor functions. However, the unit has a "stand alone" mode which eliminates the need for the terminal.

The NAT interfaces to the SCP network and a source of power through a back panel connector. The separate terminal connector is wired as Data Communication Equipment (DCE). A typical setup for development purposes would consist of an SCP standard network cable with at least three taps, a source of power (typical automotive battery voltage), the unit under test wired to the standard network, and a NAT. The NAT has the following capabilities:

- The generation of any single SCP message, triggered either from a logic-level signal or from the terminal keyboard.
- The generation of a continuous stream of messages (with programmable delays between messages) as either a single repetitive message or a programmable list of messages.
- The ability to acknowledge messages according to two user defined Lookup Tables.
- The ability to source Function Read message data as programmed by the user.
- The ability to act as a network monitor and to display SCP messages appearing on the network as hexadecimal digits. The monitor can display all messages or a subset meeting a 24-bit boolean qualification.
- The display of accumulated statistics relating to messages transmitted and received, including network fault counts.
- The ability to work in "stand alone" mode without a terminal, where it can acknowledge messages and source function read data, as well as transmit individual or multiple messages as programmed by the user.
- The ability to identify all active nodes connected to the network along with the type of interface circuit (HBCC, RAM / EBC / DUCE, core µP (MICM), etc.).

- The ability to match a given Function or Broadcast code to the nodes that would respond to that code.

- The ability to place variable data into the data fields of transmitted messages, as well as the Function Read data it sources. A voltage source (such as a potentiometer) is used to adjust the data values supplied in the message. This is intended to simulate variable data messages such as vehicle speed or coolant temperature.

- The NAT has four 32K RAM banks to extend the network monitor function. Messages may be time stamped, and the entire message buffer can be reviewed by the user via the terminal.

- The NAT can be placed in "transparent" mode, where the RS-232 port is used to provide message data to be transmitted on the network. The NAT acts as a translator from RS-232 data to the SCP network. A personal computer is assumed to provide the data to the RS-232 port.

- The NAT can provide a Trigger Output to trigger an oscilloscope upon the occurrence of a specified condition (such as network fault, message error, or predefined message).

- The NAT can accept an S-record program from the terminal port into a 32K RAM bank for execution of any given 68HC11 program.

- The NAT can accept binary data for the RAM banks from over the network for execution. The binary data can be either 68HC11 code, or a script file for message generation and acknowledgment criteria. This allows a NAT to work under the direction of a remote node on the network.

- The NAT can use the 32K RAM banks for permanent storage of alternate programs or script files.

The current hardware revision of the NAT is "D". The identifying characteristics of a revision D NAT are a small delta symbol (Δ) on the back panel and a line of text that appears above the main menu display when the NAT is reset. The current firmware revision of the NAT is 1.2. The Multiplex Design & Development section maintains a list of NAT Rev. D users who are notified of the availability of firmware upgrades when applicable.

Product Contents:

- Network Analysis Tool (NAT) Rev. D
- SCP Pigtail Cable and SCP Standard Network Cable
- NAT User's Guide

Requirements:

- RS-232 "dumb" terminal or PC with terminal emulation software
- Appropriate RS-232 cable

Available Training:

Powertrain Education offers a class that provides instruction in the basic functionality of the NAT. Contact Mike Assenmacher (313-59-41714) for further details. Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost:

$900.00 US - *Prospective users are encouraged to verify cost and availability prior to ordering.*

2.2 Dual SCP Peripheral Adaptor (Dual SPA / DSPA)

The Dual SPA allows an IBM-PC/XT/AT backplane compatible computer to interface to an SCP network. The Dual SPA is a half-length, Industry Standard Architecture (ISA) bus plug-in card that essentially provides two independently configurable HBCCs controlled by a 16 MHz 80C196. A PC equipped with a Dual SPA card can perform the network operations of transmission, reception, and monitoring. The Dual SPA was designed to be a high speed device appropriate for network development or test. The capabilities of the Dual SPA card are much the same as the NAT with the following additional features or enhancements:

- Incorporation of a 64 message receive buffer per HBCC.
- Incorporation of a 16 message transmit buffer per HBCC.
- Six auto-transmit messages (per HBCC) triggered by a user-defined event.
- Four user-defined filters (per HBCC) that control the actions associated with a received SCP message.
- Additional A/D channels.
- Additional trigger in and trigger out capabilities.
- The ability to log data to a PC file for post-processing and analysis of network data.
- The ability to create custom applications using the included primitive-level libraries).
- Supports up to four (4) installed DSPA cards per PC.

Product Contents:

- Dual SPA Card
- SCP Pigtail Cable and SCP Standard Network Cable
- Primitive-level Libraries (Borland® and Microsoft®)
- Sample Interface Software
- Example Source Code
- Dual SCP User's Guide and Primitive-level Library Interface Specification

Requirements:

- IBM® PC/XT/AT or compatible
- 4K of free RAM above 640K (memory range D000-F000) for each Dual SPA card installed

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost:

$1000.00 US - *Prospective users are encouraged to verify cost and availability prior to ordering.*

2.3 Network Simulator

This Microsoft® Windows® based software allows modeling of SCP nodes and their associated message traffic as specified by an ASCII file generated by the Vehicle Configuration Editor (VCE).. Information is produced by the simulator regarding message latency, arbitration losses, and node querying, along with an optional log file of events. This tool is normally provided as part of the Multiplex Development System.

Note: This tool is only provided to those engineers that can demonstrate a need for its use.

Product Contents:

- Network Simulator Software
- User Documentation
- *Part of the Multiplex Development System*

Requirements:

- Microsoft® Windows® 3.x
- 1 MB Hard Disk Space

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost: Free

2.4 Data Logger

The Data Logger is a hardware / software pair that is used to monitor vehicle network traffic. The Data Logger is a 60-pin compatible module that is placed in a vehicle (usually in the trunk) and connected to the SCP network. The Data Logger then monitors SCP network traffic and records messages and calculates statistics regarding the general performance of the network (e.g., # of messages, # of bits, utilization). The Data Logger software (DLLink) is used to interface to the Data Logger for configuration, uploading of data, and re-programming of firmware. Furthermore, the Data Logger software can be used to configure the Data Logger to filter specific messages so that only a pre-defined set are considered.

The Data Logger can be used for either SCP or ACP vehicle networks.

Specific functions of the Data Logger include:

- Capturing network violations (e.g., invalid checksum, invalid messages, wrong acknowledgers, etc.) and network statistics (e.g., # messages received, etc.)

- Triggering message transmissions based on specified events such as reception of messages or specified time delays

- Filtering of specified messages based on a user specified condition (e.g., Target = Fx and Data Byte #1 < > 22)

- Logging or transmitting messages based on hardware signals such as Trigger-in, Ignition Key On (ASYS-ON), or Ignition Key Off (ASYS-OFF).

In addition, the Data Logger can be configured either in stand-alone mode in which all data is captured and then reviewed later via the Data Retriever or in "real-time" mode in which specific data is displayed in "real-time" to the user via the Data Retriever.

For more information, please consult the Data Logger / Data Retriever User's Guide.

Product Contents:
- Data Logger
- Data Retriever
- User Documentation

Requirements:
- IBM® PC/XT/AT or compatible
- Dual SPA Card

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost:

$1000.00 US - *Prospective users are encouraged to verify cost and availability prior to ordering.*

Note: The SCP and ACP Data Logger is one device.

2.5 ECU Test Tool

The ECU Test Tool is a Microsoft® Windows® based software product that allows a user to perform tasks common to module development and testing. Several functions are provided such as network monitoring, PID survey and monitoring, memory monitoring, DTC survey and monitoring, diagnostic command survey and monitoring, diagnostic routine survey and monitoring, and data recording. In addition, any valid SCP message may be defined and transmitted.

Version 3.0 will include several enhancements including: interface to the Multiplex Reference DataBase (MRDB), Macros, Message Filtering, and SCP HBCC Configuration.

Product Contents:

- ECU Test Tool
- User Documentation

Requirements:

- IBM® PC/XT/AT or compatible
- Dual SPA Card
- Microsoft® Windows® 3.1
- 5 MB hard disk Space

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost: Free

2.6 Message Verification Test Tool

The Message Verification Test Tool is a DOS-based product designed to provide vehicle-level SCP network verification. Tests include: Node Continuity, PID Survey, Primary ID Test, Function Read Test, Status Request Test, Periodic Message Test, Initialization Test, Periodic Event Test, and Event Test. The test tool is driven by an ASCII file that completely defines an SCP vehicle network. This ASCII input file can be generated by hand or automatically from the Vehicle Configuration Editor (VCE).

The test tool derives the appropriate tests based on the input file and generates a result file. This tool is used by E/ESE - Multiplex Standards & Application Section as the primary method of SCP vehicle network verification.

Product Contents:

- Message Verification Test Tool
- User Documentation

Requirements:

- IBM® PC/XT/AT or compatible
- Dual SPA Card
- Appropriate Vehicle Input File
- 1 MB hard disk Space

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost: Free

3.0 ISO-9141 Products

Several products are available which apply specifically to ISO-9141 networks. These tools allow a user to develop, test, diagnose, and simulate modules on an ISO-9141 network. This section details the following tools:

- Multicom II Interface Card
- ECU Test Tool

These tools and their capabilities are presented below in more detail.

3.1 MultiCom II

The MultiCom II is a half-length Industry Standard Architecture (ISA) plug-in card which allows an IBM® PC/XT/AT backplane compatible computer to interface to an ISO-9141 network. A PC containing a MultiCom II card can perform any network activities such as transmit, receive, and monitor. Furthermore, up to four MultiCom II cards may be installed in a single PC.

The MultiCom II provides several advanced features including: message receive and transmit buffers, A/D channels, trigger in and out capabilities, ISO-9141 stream mode, normal mode and European mode. The MultiCom II is controlled through a set of "C" primitive-level functions which allow complete control of the board.

Product Contents:

- MultiCom II card
- Primitive-Level Libraries (Borland® and Microsoft®)
- Simple Interface Software
- Example Source Code
- User Documentation

Requirements:

- IBM® PC/XT/AT or compatible
- 4K of free RAM above 640K (memory range D000-F000) for each MultiCom II card installed

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost:

$1150.00 US - *Prospective users are encouraged to verify cost and availability prior to ordering.*

3.2 ECU Test Tool

The ECU Test Tool is a Microsoft® Windows-based software product that allows a user to perform tasks common to module development and testing. Several function are provided such as network monitoring, PID Survey and monitoring, memory monitoring, DTC Survey and Monitoring, diagnostic command survey and monitoring, diagnostic routine survey and monitoring, and data recording. In addition, any ISO-9141 message may be defined and transmitted. Note that this is the same tool described in section 2.6.

Version 3.0 will include several enhancements including: interface to the Multiplex Reference DataBase (MRDB), Macros, and Message Filtering.

Product Contents:
- ECU Test Tool
- User Documentation

Requirements:
- IBM PC/XT/AT or compatible
- MultiCom II Card
- Microsoft® Windows 3.1
- 5 MB hard disk Space

Available Training:
Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost: Free

4.0 ACP Products

Several products are available which apply to ACP networks. These tools allow a user to develop, test, diagnose, and simulate modules on an ACP network. This section details the following tools:

- Data Logger
- Message Verification Test Tool

These tools and their capabilities are presented below in more detail.

4.1 Data Logger / Data Retriever

See section 2.4 Data Logger for specific information. The SCP and ACP data logger is one device.

4.2 Message Verification Test Tool

The Message Verification Test Tool is a DOS-based product designed to provide vehicle-level ACP network verification. Tests include a Message Conversation Test, Periodic Message Test, Initialization Test, Periodic Event Test, and Event Test. The test tool is driven by an ASCII file which completely defines an ACP vehicle network. The test tool derives the appropriate tests based on the input file and generates a result file. This tool is used by E/ESE - Multiplex Design & Development Section as the primary method of ACP vehicle network verification.

Product Contents:

- Message Verification Test Tool
- User Documentation

Requirements:

- IBM PC/XT/AT or compatible
- ACP Network Analyzer Tool
- Appropriate Vehicle Input File
- 1 MB hard disk Space

Available Training:

Contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS..

Unit Cost: Free

5.0 Procurement and Upgrade Procedure

For information regarding pricing and tool availability contact the network tools hotline @ 32-25870 or send a message to PROFS: NETTOOLS.

Appendix D ASSOCIATED DOCUMENTATION

D.1 PURPOSE.

This Appendix provides a list of the documents that are either referenced by this specification or can provide detailed information about material covered within this specification.

Unless specified otherwise, the applicable revision level for the following documents are those in effect at the time of release of this specification.

D.2 FORD DOCUMENTATION

The following documentation has been developed within the Advanced Vehicle Technology, Core & Advanced Electrical / Electronics Systems Engineering Office or other FORD organizations. For assistance in attaining any of these documents, contact the AVT C&A EESE Office.

1.	FORD-9141 Protocol Definitions and Interface Requirements	F7LC-12K529-FD
2.	SCP Protocol Definition and Interface Requirements (SCP-001)	F7LC-12K529-BC
3.	SCP Vehicle Network Implementation Requirements (SCP-003)	F7LC-12K529-DE
4.	Module Programming/Configuration Design & Process Specification	ES-XW4T-1A294-CA
5.	Worldwide Customer Requirements for Passenger Cars	Manual No. 905
6.	Worldwide Customer Requirements for Light Truck	Manual No. 506

D.3 NON-FORD DOCUMENTATION

The following documentation has not been developed within FORD. For assistance in attaining any of the following documents, contact the SAE or the AVT C&A EESE Office.

1.	Recommended Practice for Class "B" Data Communications Network Interface	SAE J1850
2.	Recommended Practice for Diagnostic Connector	SAE J1962
3.	Recommended Practice for Diagnostic Test Modes	SAE J1979
4.	Recommended Practice for Diagnostic Trouble Code Definitions	SAE J2012
5.	Recommended Practice for Data Formats	SAE J2178
6.	Recommended Practice for Enhanced E/E Diagnostic Test Modes	SAE J2190
7.	Glossary of Vehicle Networks for Multiplexing and Data Communications	SAE J1213
8.	OBD II Scan Tool	SAE J1978
9.	Security Access	SAE J2186

10. Module Programming/Configuration Design & Process Specification ES-XW4T-1A294-CA

Appendix E DEFINITIONS

The definitions provided in SAE J1213 Glossary apply to this specification. Additional or modified definitions of specific terms unique to this document are listed below:

CARB	**C**alifornia **A**ir **R**esources **B**oard - administrative group that formulates recommendations for emission regulation in California.
DMR	**D**irect **M**emory **R**ead - A method of obtaining information from an ECU by accessing its memory address space.
ECU	**E**lectronic **C**ontrol **U**nit - An onboard electronic module that controls a vehicle subsystem function.
EPA	**E**nvironmental **P**rotection **A**gency (Federal Agency).
OBDII	Set of emissions related stipulations that are being required by California Air Resources Board.
Offset	A number used to refer to a data value by specifying its relative position in a list of data values.
PID	**P**arameter **ID**entifier - an index number used to refer to a parameter within a module without knowing its storage location.
SCP	Generic reference to the **S**tandard **C**orporate **P**rotocol for multiplex communication.
SCP-DIAG	Reference to the specific communication methodology that enhances diagnostic and manufacturing capability by use of the SCP communication bus.
Start of Frame Marker	Unique delimiter that signifies the start of a frame. This marker is used by all SCP modules to synchronize internal timers.
Stored Codes	Also referred to as Fault Codes, Diagnostic Trouble Codes or DTC's.
Tester	Any offboard device that may cause an ECU to enter a test mode or communicate stored codes, PIDs or status.